

PROJECT OVERVIEW

AI-Powered BI — Upload → Analyse → Ask

Excel/CSV upload → auto clean → smart dashboard → Gemini Q&A chatbot

BUILD PHASES

Phase 1 · Week 1 **File upload + data cleaning engine**

- Accept .xlsx, .xls, .csv via drag & drop
- Auto-detect column types (numeric, date, category, text)
- Handle missing values, duplicates, messy headers
- **Data cleaning weight: medium** — Pandas covers 90% of it; heavy only if user uploads truly chaotic data
- Show a "data health" summary card (nulls%, duplicates, outliers)

Phase 2 · Week 2 **AI-driven auto-visualization**

- Gemini API reads column names + sample data → decides best chart types
- Auto-renders: bar, line, pie, scatter, heatmap, histogram — no user input
- Gemini writes a 3-line plain-English insight under each chart
- Anomaly detection callout: "Sales on 14 Jan were 3× the monthly average"
- Dashboard layout auto-arranged (KPI cards at top, charts below)

Phase 3 · Week 3 **Natural language Q&A chatbot**

- Sidebar chat: "Which product had highest return rate?"
- Gemini gets the question + dataframe summary as context → answers in plain English
- If question needs a chart, generate it on the fly and show inline in chat
- Conversation memory (multi-turn: "now filter that by Q3")

Phase 4 · Week 4 **Polish + deploy**

- Export dashboard as PDF or PNG
- Share a public link to your dashboard (session-based)
- Multi-file merge (join two sheets on a common column)
- Deploy to Streamlit Cloud (free) + Render free tier as backup

TECH STACK

Frontend / App
Streamlit

Data engine
Pandas + NumPy

Charts
Plotly Express

Excel reading
openpyxl

TECH STACK

Frontend / App Streamlit	Data engine Pandas + NumPy	Charts Plotly Express	Excel reading openpyxl
AI brain Gemini 1.5 Flash	Gemini SDK google-generativeai	PDF export kaleido + reportlab	Deploy Streamlit Cloud

MARKET DIFFERENTIATORS (STAY AHEAD)

-  **Auto anomaly narrator** — not just a red dot on a chart, Gemini writes a sentence explaining *why* it's anomalous using surrounding data context
-  **Column relationship discovery** — Gemini auto-finds correlations you didn't ask for: "Revenue and ad spend are 0.87 correlated"
-  **Business-mode summaries** — one-click "write me an executive summary of this data" as a formatted report
-  **Multi-sheet merge Q&A** — upload 2 Excel files, ask questions across both joined on a key column
-  **Forecast tab** — Gemini + simple statsmodels linear trend for "predicted next 30 days" on time-series data

FREE DEPLOYMENT PLAN

-  Streamlit Cloud — primary (free, GitHub-connected)
-  Render free tier — backup
-  Gemini API — free via Jio AI+ plan
-  GitHub — source + CI